The Confidence Paradox

Confidence tracks correctness within people more consistently than average confidence distinguishes between people.

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KEY RESULT | THREE PERCEPTUAL-DECISION DATASETS

In three selected visual tasks, logistic population-averaged GEE models found positive within-person confidence–accuracy associations. All three corresponding between-person 95% robust confidence intervals spanned zero. These are exploratory observational associations, not causal findings or a prospective preregistration.

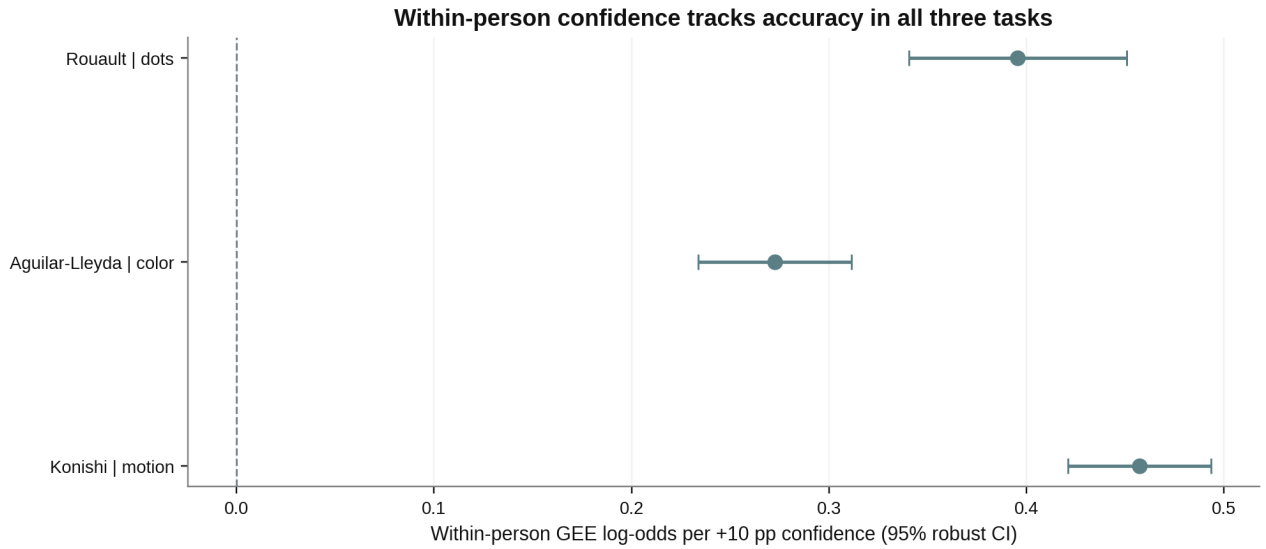


Figure 1. Each point is a GEE log-odds coefficient for a +10 percentage-point rise in confidence relative to the same individual's mean; whiskers are participant-cluster robust 95% confidence intervals.

Research question

Is an individual more likely to be correct when unusually confident in a particular answer, and is this distinct from the relationship between different individuals' average confidence and accuracy?

Scope and contribution

We re-analyze open perceptual-decision data from Rouault, Aguilar-Lleyda and Konishi. The contribution is a transparent separation of within-person and between-person levels, with clustering, sensitivity checks and a reproducible pipeline—not a claim that the confidence–accuracy relationship is newly discovered.

Scientific status: independent student exploratory re-analysis; existing data; hypotheses were not publicly preregistered; not peer-reviewed. Analytical coding and report preparation included AI assistance, with reproducible source and documented limitations.

1 Data and methods

Data origin

All three CSVs and their contributor READMEs originate in The Confidence Database (Rahnev et al., 2020; OSF: https://osf.io/s46pr/). Each decision has objective correctness coded 0/1; reported confidence is 50–100% (Rouault stores its equivalent as 0.5–1.0). No new participants were recruited for this project.

Primary study	People	Rated decisions	Primary task
Rouault 2019	72	12,960	Dot numerosity
Aguilar-Lleyda 2018	28	11,200	Color discrimination
Konishi 2019	93	22,320	Single-task motion

Model specification

The primary models use logistic GEE with participant-cluster robust standard errors. We enter trial confidence minus participant mean (within-person) and participant mean minus grand mean (between-person), each measured in 10 percentage-point units, with an adjustment for trial difficulty. Rouault and Aguilar-Lleyda use exchangeable working correlation; Konishi uses independence working correlation with robust clustering and experimental-condition fixed effects.

The unit of independent clustering is the person, not the repeated decision. Konishi contains repeat study sessions for some participants; they retain the same person ID. Aguilar-Lleyda records color and letter choices in the same trial: primary color is analyzed alone and letter is supplementary. No task or subject is treated as randomly assigned to another study.

Sensitivity and uncertainty

Rouault includes exploratory participant-quality exclusions because the original article’s final 46-person selection cannot be fully reconstructed from public comprehension-question data; 72 people with confidence reports enter this re-analysis. Konishi within-session person-mean centering was also checked. Aguilar-Lleyda adds a participant-cluster bootstrap because only 28 participants contribute to that study; such small-cluster inference is still exploratory.

Rouault additionally compares GEE to a conditional random-intercept logistic GLMM fitted by adaptive Gauss-Hermite marginal likelihood and resampling complete participant clusters. This method replaces the previous, over-precise variational-Bayes interval; the two model families target different marginal/conditional associations.

Interpretive boundaries

A positive coefficient predicts larger odds of correctness, not an equal change in the probability or an effect of manipulating confidence. A bin of average confidence versus average accuracy illustrates average calibration but does not estimate a trial-level calibration curve or metacognitive sensitivity in the signal-detection sense.

2 Results

Primary associations

Primary task	Within (95% robust CI)	Between (95% robust CI)
Rouault: Dot numerosity	+0.396 [+0.341, +0.451]	+0.046 [-0.031, +0.124]
Aguilar-Lleyda: Color discrimination	+0.273 [+0.234, +0.311]	+0.037 [-0.037, +0.111]
Konishi: Single-task motion	+0.457 [+0.421, +0.493]	-0.055 [-0.169, +0.059]

All effects are GEE log-odds slopes per +10 percentage points confidence. Their intervals are not independent across primary analyses if studies share participants; no formal cross-study meta-analysis was fitted. All three within-person intervals are positive; all three between-person GEE intervals include zero.

Within-minus-between contrasts

Exploratory joint-model contrasts (within slope minus between slope): Rouault +0.349 [+0.255, +0.443]; Aguilar-Lleyda (color) +0.236 [+0.138, +0.333]; Konishi (single motion) +0.513 [+0.384, +0.641]. These are model-based associations and involve different tasks and participants.

Sensitivity checks

Rouault exploratory screening retained 72, 71, 59, 54 people across four conditions, and within-person GEE estimates ranged from +0.396 to +0.442. In Konishi, centering within each participant–experiment session yielded a within slope of +0.456 [+0.421, +0.491], compared with +0.457 under participant-wide centering.

For Aguilar-Lleyda, a 298-draw participant-cluster percentile bootstrap produced a color-task within-person interval [+0.233, +0.322]. With only 28 independent people, confidence limits should not be regarded as conclusive without additional samples.

Between-person GLMM versus GEE: unresolved inferential sensitivity

The previous variational-Bayes interval was judged unreliable and withdrawn as evidence. In Rouault, GEE gave between-person +0.046 [-0.031, +0.124] (marginal), while the numerically integrated GLMM gave +0.151 [+0.002, +0.297] (conditional; 500 participant-bootstrap draws). The GLMM lower limit lies very close to zero; conclusions about the between-person effect are method-sensitive and fragile. These two model coefficients are not directly interchangeable.

3 Interpretation and limitations

What can be concluded?

Within the examined perceptual tasks, a person tends to be more accurate when their confidence exceeds their usual level. This pattern was seen in three selected sources under prespecified-in-code but not preregistered analysis choices. It does not establish an intervention effect of increasing confidence, a general human personality trait, or a result extending to economic, emotional, clinical or human–AI decisions.

Limitations a PI should inspect

• Only selected visual perceptual tasks were studied, and the sample-selection rationale evolved during exploration. Konishi was held aside during development but there was no formal preregistration.

• Rouault original exclusion rules selected 46 of 73; full comprehension-question data are absent, so this archive uses alternative exploratory screens.

• Between-person inference uses relatively few independent clusters compared with total decisions; the Aguilar-Lleyda participant bootstrap is sensitive to the limited 28-person pool.

• Difficulty is adjusted, but trial order, fatigue, learning and differences in feedback or experimental setting may still confound associations.

• Multiple models, outcomes and sensitivity checks create multiplicity concerns; results should not be read as a familywise-confirmatory discovery.

• The earlier variational-Bayes model gave a suspiciously narrow interval. Numerical GLMM refitting lessens but does not remove model dependence; calibration and discrimination remain distinct outcomes.

Reproduce the analysis

From the supplied Python environment install dependencies with pip install -r requirements.txt, then execute python run_all.py. The runner rebuilds CSV results, plots and this report, then calls validation tests. To inspect the corrected mixed-model conflict, see docs/STATISTICAL_AUDIT.md. The source CSVs are unchanged and are available under the Confidence Database CC0 data release; original source studies must be credited.

Research transparency

This is a student secondary-data project prepared with AI coding assistance. It is not a peer-reviewed publication. A standalone prospective preregistered study, robust replication with new participants, and external-domain data would be needed for stronger generalizability claims.

References

• Rahnev, D., et al. (2020). The Confidence Database. Nature Human Behaviour 4:317–325. doi:10.1038/s41562-019-0813-1.

• Rouault, M., Dayan, P., & Fleming, S. M. (2019). Forming global beliefs about self-performance from local confidence. Nature Communications 10:1141. doi:10.1038/s41467-019-09075-3.

• Aguilar-Lleyda, D., Lemarchand, M., & de Gardelle, V. (2018). Confidence as a priority signal. bioRxiv 480350. (See original README.)

• Konishi, M., et al. (2019). Metacognition is spared in a dual-task visual paradigm. doi:10.31234/osf.io/xcseh.

• Data: https://osf.io/s46pr/; each study’s contributed README is retained in the archive.